CMPE 110 Fall 2025
Monday 4:30-7:15 PM
Lab #2: Second-Order Passive Circuits
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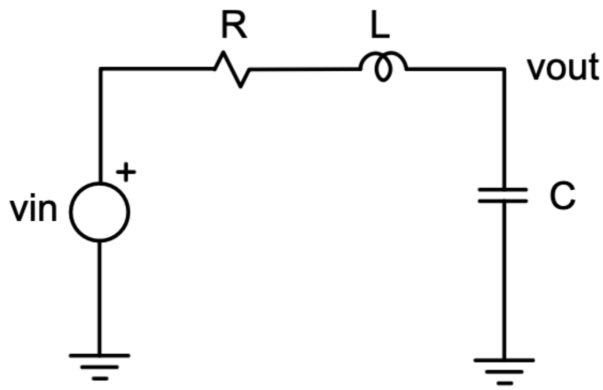
Date: 09/20/25

PART 1: RC Circuits



$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

When $X_L = X_C$ resonance occurs

$$f_o = \frac{1}{2\pi\sqrt{LC}}$$

b.) $a = \frac{R}{2L}$

$s_1 = -a + \sqrt{a^2 - \omega_o^2}$

$s_2 = -a - \sqrt{a^2 - \omega_o^2}$

Damping ratio $\zeta = \frac{R}{2\sqrt{L/C}}$

Q-Factor for Series RLC: $\frac{\sqrt{L/C}}{R}$

A) Calculate L so that the oscillation angular frequency, w0, is 10e7 rad/sec if C=100pF is used.
   a) L=100uH

a.) $\omega_o = 2\pi f_o = \frac{1}{\sqrt{LC}}$

$\sqrt{LC} = \frac{1}{\omega_o}$

$LC = \frac{1}{\omega_o^2}$

$L = \frac{1}{\omega_o^2 C} = \frac{1}{(10^7)^2 \cdot 100^{-12}}$

$= \frac{1}{10^4} = 10^{-4}\,H$

$= 100\ \mu H$

B) Compute the poles for each RLC combination (j=imaginary component)

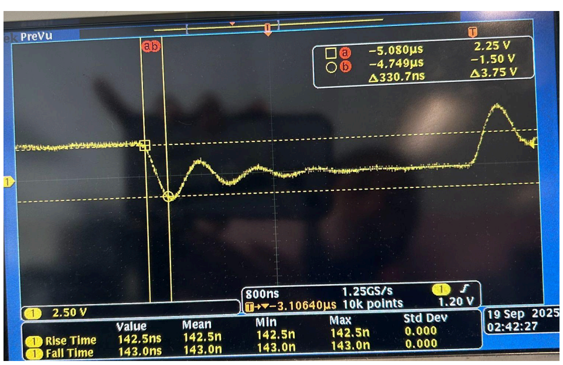
Poles of RLC, L=100uH, C=100pF, w0=10^7			
R (Ω)	**a=R/(2L)**	**s1 (a+jb)**	**s2 (a+jb)**
0	0.00E+00	+je7	-je7

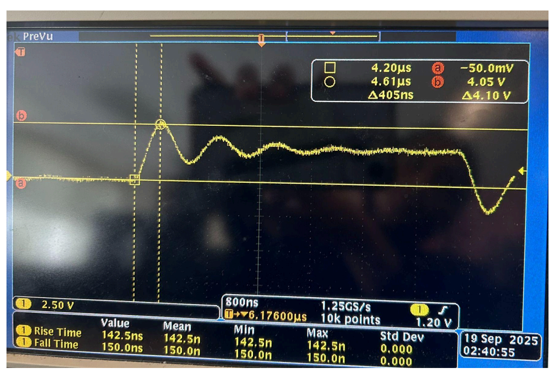
10	5.00E+04	-5*e4+j9.99e7	-5*e4-j9.99e7
100	5.00E+05	-5e5+j9.99e6	-5e5-j9.99e6
1000	5.00E+06	-5e6+j8.66e6	5e6-j8.66e6
10000	5.00E+07	-1.01E+06	-4.9E+07

C) [1]Apply pulse generator at Vin that changes between 0-5V and measure rise and fall times at Vout for each RLC circuit in part B.

Actual vs Expected Rise and Fall Times (0-5V pulse)						
R (Ω)	**meas_rise (s)**	**meas_fall (s)**	**spice_rise (s)**	**spice_fall (s)**	**%error_rise**	**%error_fall**
0	1.39E-07	1.47E-07	1.65E-07	1.58E-07	-1.57E+01	-7.09E+00
10	1.37E-07	1.47E-07	1.59E-07	1.57E-07	-13.53904282	-6.5605096
100	1.43E-07	1.44E-07	1.67E-07	1.61E-07	-14.39712058	-10.559006
1000	3.51E-07	3.63E-07	2.48E-07	2.56E-07	41.41129032	41.6796875
10000	5.07E-07	4.92E-07	2.14E-06	2.18E-06	-76.30841121	-77.431193

[2]Waveform of the Circuit With a Resistor Value of 0Ohm

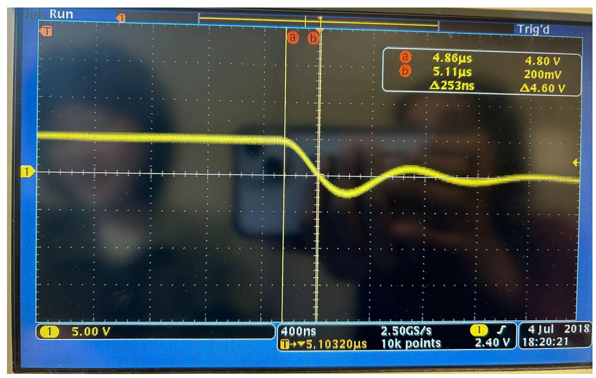


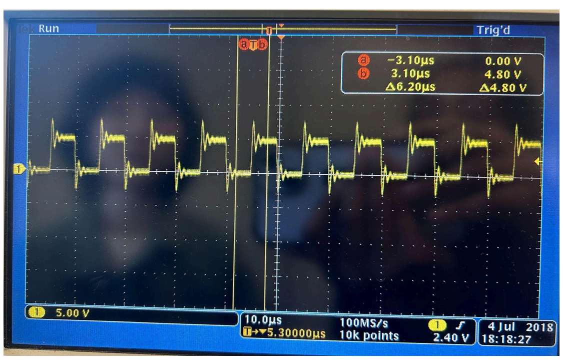


[1] For rise times, we assume the percentile ranges (0-100% of final value) for under damped circuits, as stated in lecture, and 10-90% for overdamped cases. For underdamped RLC circuits, we measure the rise and fall time based on the decaying sine wave, not its exponential envelope.
[2] *rise time of underdamped circuit is 0-100%

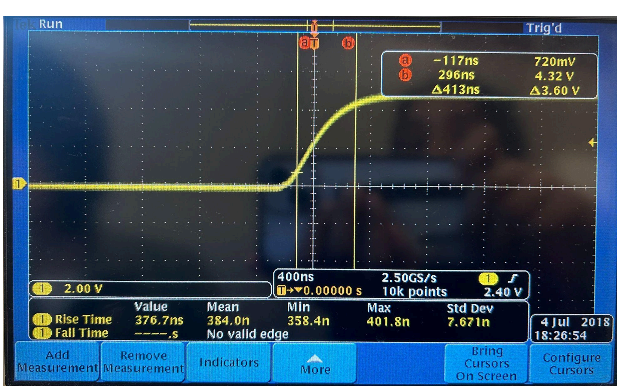
Waveform of the Circuit With a Resistor Value of 10Ohm

Waveform of the Circuit With a Resistor Value of 100Ohm





Waveform of the Circuit With a Resistor Value of 1000Ohm



Waveform of the Circuit With a Resistor Value of 10000Ohm

D) Apply square-wave generator at Vin that changes between 0-5V and measure the time constant (if applicable) at Vout for each RLC circuit.

Actual vs Expected Time Constant (0-5V square)			
R (Ω)	**spice_tau (s)**	**meas_tau (s)**	**%error**
0	N/A	N/A	N/A
10	2.00E-05	1.92E-07	-9.90E+01
100	2.00E-06	1.92E-07	-9.04E+01
1000	2.00E-07	2.93E-07	4.65E+01
10000	1.00E-06	2.01E-06	1.01E+02

i) [3]Underdamped ($\zeta < 1$) : $\tau = 2L/R$
ii) Critically damped ($\zeta = 1$) : $\tau = 2L/R$
iii) Overdamped ($\zeta > 1$): $\tau = RC$

[4]Time Constant of the Circuit With a Resistor Value of 0Ohm

E) Apply pulse generator at Vin that changes between 0-5V and measure the free oscillation frequency (resonant frequency of undamped LC circuit), f0=w0/ 2pi, at Vout for each RLC circuit.

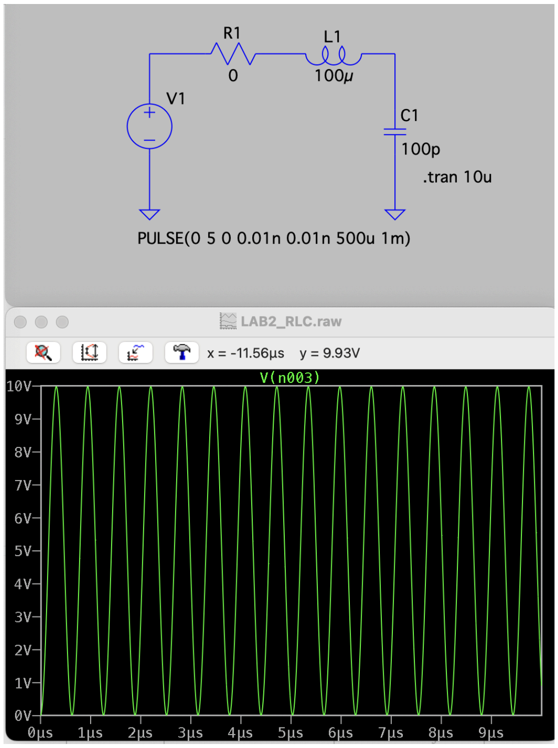
a) Calculated formula: $f0 = \frac{1}{2\pi\sqrt{LC}}$

Actual vs Expected Free Oscillation Freq (0-5V pulse)			
R (Ω)	**calc_f0 (Hz)**	**meas_f0 (Hz)**	**%error**
0	1.59E+06	1.00E+06	-37.16814693
10	1.59E+06	9.90E+05	-37.79024511
100	1.59E+06	8.00E+05	-49.73451754
1000	1.59E+06	8.00E+05	-49.73451754
10000	N/A	N/A	N/A

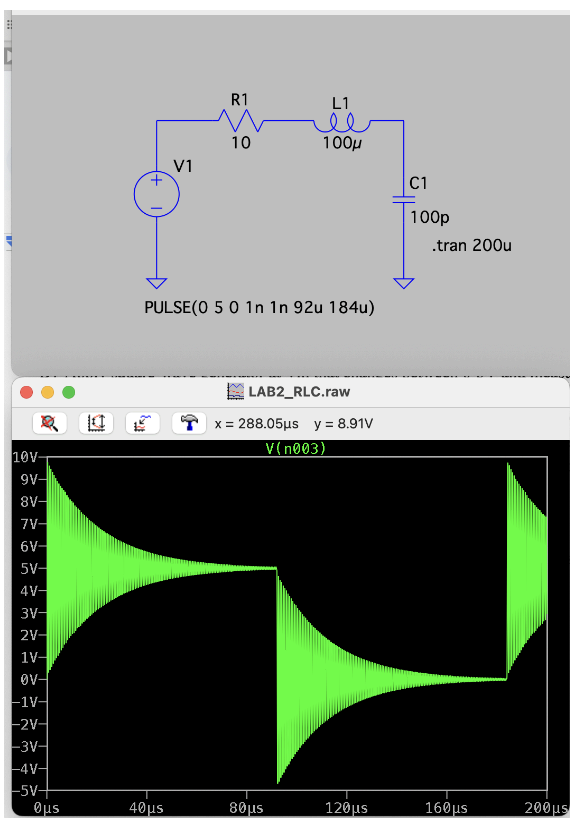
F) [5]Error analysis: simulate all cases in LTSpice and compare results.

[3] The time constant for RLC circuits is not as straightforward to measure as with first order circuits due to oscillation. Since the time constant represents the time it takes for the amplitude of the oscillation to rise to about 63% of the initial Vin value, we will be measuring it on the envelope of the waveform for underdamped circuits.
[4] *measure time constant by measuring from 0 to 63%, so 0-3.15V
[5] See tables above for percent error

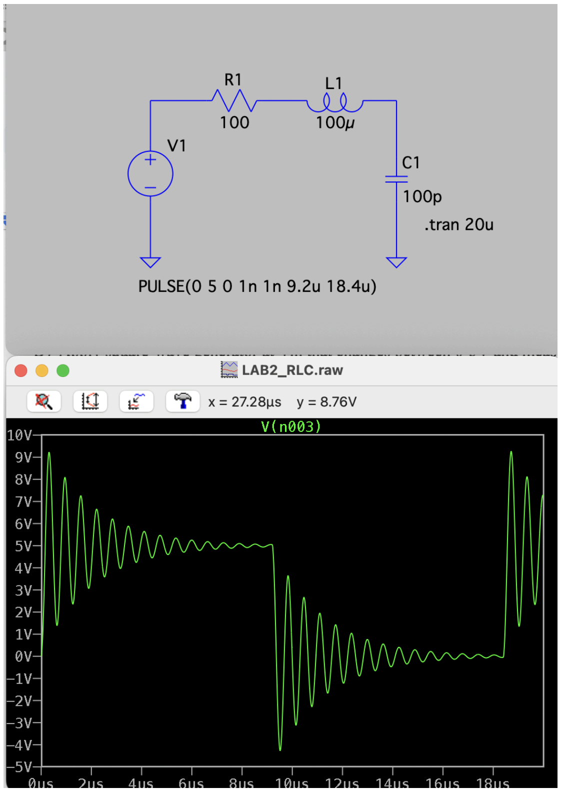
LTSpice Waveform of the Circuit With a Resistor Value of 0Ohm



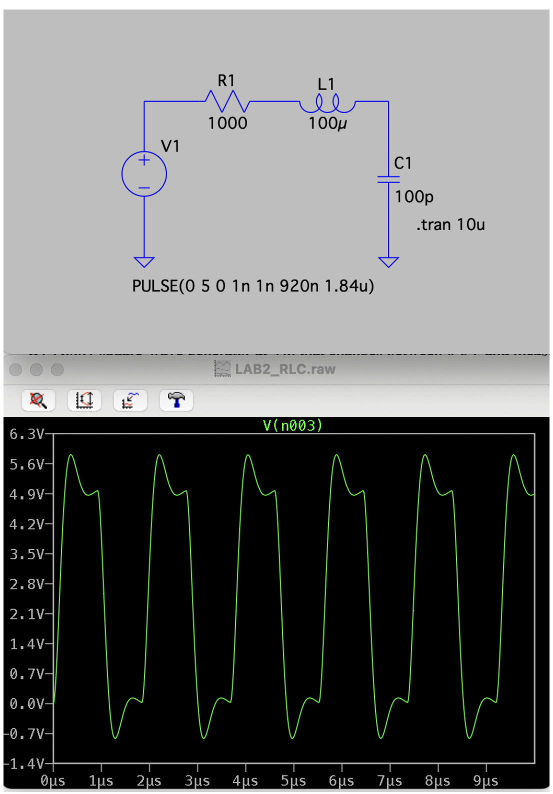
LTSpice Waveform of the Circuit With a Resistor Value of 10Ohm



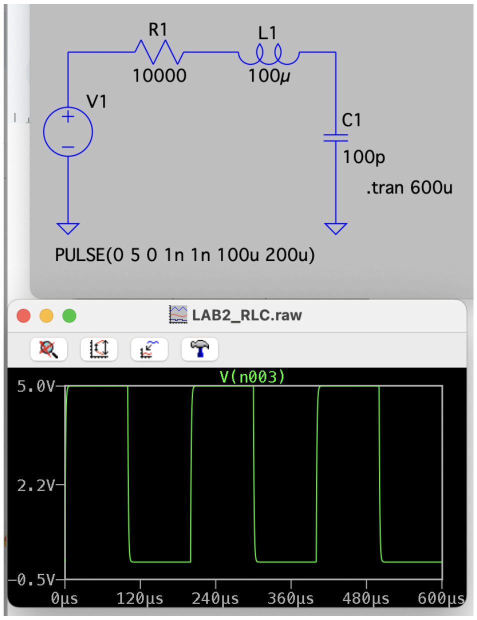
LTSpice Waveform of the Circuit With a Resistor Value of 100Ohm



LTSpice Waveform of the Circuit With a Resistor Value of 1000Ohm



LTSpice Waveform of the Circuit With a Resistor Value of 10000Ohm



Conclusion:

In this lab we studied the response of RLC circuits with the same inductor but varying resistor values. We studied the output waveforms to determine the time constants, rise and fall times. We determined that all of the circuits were underdamped except for the last one, which was overdamped. For underdamped circuits, we found the rise and fall times by measuring the time it takes to get from 0% to 100% of the final voltage, which is 0V to 5V in this lab. For overdamped circuits, we measured from 10% to 90% of the final voltage to get the rise and fall times. We found the time constant by measuring the time it takes to get from 0% to 63% of the final voltage. We built these circuits on a breadboard as well as on LTSpice and compared the results. While the values are quite similar, some of the measured values from the oscilloscope are greater than the expected values from the simulation. This may be caused by parasitic capacitances or inductances from the equipment.